

Lecture Notes on

Quantum Computing

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1 9/23/24 EPR Paradox and Bell's Theorem

Standard and diagonal basis are incompatible or complementary

Proof of EPR Pair Suppose for contradiction that such a state $|\psi\rangle = |a\rangle \otimes |b\rangle$. We can rewrite this in the standard basis and we see through contradiction that states $|a_0a_b\rangle = 1/\sqrt{2}, |a_1b_1\rangle = 1/\sqrt{2}$ and the other cases are 0. Thus, we see a contradiction in the state for the EPR pairs.

EPR Paradox Suppose we have Alice and Bob with the state $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Consider two experiments where Alice measures her qubits in the standard basis and in the other one, she measures in the diagonal basis.

In the standard basis, it is intuitive we get $|0\rangle$ 50% of the time and so for $|1\rangle$. To find the result in experiment 2, we have that

$$|v_+\rangle = (\langle + | \otimes I) |\phi\rangle = \frac{1}{\sqrt{2}}(\langle + | 0 \rangle |0\rangle + \langle + | 1 \rangle |1\rangle) = \frac{1}{2}(|0\rangle + |1\rangle)$$

Normalizing we will get that the probability of Alice getting $|+\rangle = \frac{1}{2}$. The post measurement state is then $|+\rangle \otimes |+\rangle = \frac{1}{2}$. Therefore no matter the choice, we will get the same probability w.r.t any basis.

CHSH Game Consider a hidden variable that could possibly help maximize the probability from 3/4. However, we realize that it is impossible. However, there exists a quantum strategy that uses entangled qubits with a win rate of 85.4%. Therefore, this must rule out the local hidden variable theorem.

Alice will measure in the standard bases and Bob will measure in the rotated standard basis by $\pi/8$ depending on the answer of y.

Now supposed $x = 0, y = 0$. We can analyze what happens and it doesn't matter if Alice measures first or Bob measures first. If Alice measures first, we have that $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$. Bob now measures his qubit and gets some outcome $|s_0\rangle$ where he maps to 0. We see that $|\langle 0 | s_0 \rangle|^2 = 0.85$. We can check that with s_1 , we have $|s_1\rangle \rightarrow 1$ which is 0.15. This is true for both $|00\rangle$ and $|11\rangle$.

We now consider Alice measuring in the $|+\rangle$ basis. We now have Bob measure in the $|t_0\rangle$ basis which will get us the same result of 0.85 based on the inner product.

2 9/30/24 Holevo's Theorem and Introduction to Quantum Computation

Can we use n qubits to store classical information such that we can store more information than n bits. This is not possible due to Holevo's Theorem

Consider the following: Alice has an m bit string X and she wants to transmit this information to Bob. She decides to send quantum information such that Bob can perform some computation to decode X . This is only possible if the number of qubits is at least m where m is an arbitrary number.

This is possible to do, however we need Alice to preshare an EPR pair which gives rise to superdense coding. The total number of EPR pairs is $m/2$ where 1ebit + 1qubit = 2cbits.

2.1 Gate Sets

When considering the gate sets used for a given hardware system. We will consider the gate set $\mathcal{G}$ as the universal gate set where for any unitary U , one can construct any circuit using gates from $\mathcal{G}$. Additionally, we must understand how unitaries approximate one another. We will say that unitary U approximates state V if $\|U|\psi\rangle - V|\psi\rangle\| < \epsilon$. We can also create circuits that approximate a certain unitary. That is, we do not measure the circuit but take the final result as some existing unitary. Therefore, we can say a circuit C can approximate a unitary.

We will show the following continuous gate set is universal $R_X(\theta), R_Y(\theta), R_Z(\theta), P(\varphi)$ where these gates have parameters that can be shifted. A discrete finite universal gate set could be $H, S, T, CNOT$. Note, the discrete gate set is interesting in the sense that we can approximate any unitary to a degree less than ϵ . Now for particular gate sets, we wish that the compilation will not use an excessive amount of gates. Therefore we wish that C' will be minimized to a certain amount.

To prove this, we can use the Solovay-Kitaev Theorem to deduce that this.

Solovay-Kitaev Theorem Let $\Gamma \subseteq SU(2)$ (Special Unitary Group of Unit 2) which is the set of all single qubit unitaries up to a global phase.

1. Γ generates a dense subgroup of $SU(2)$ where the set of all Γ will stay in the ball
2. Γ is closed on its inverse, which means that we can find the inverse which is trivially true for unitary gates.

Now if we have this set Γ , the gate $U \in SU(2), U \notin \Gamma$ can be approximated to ϵ in at most $\mathcal{O}(\log(\frac{1}{\epsilon}))$

Corollary For a gate set satisfying the Solovay-Katev Theorem. Then for all n and circuits C and for all $\epsilon > 0$, there exists a circuit C' consisting of gates from $\Gamma \cup CNOT$ where ϵ approximates C .

This is called quantum circuit synthesis. That is, how many single and two qubit gates are needed to build a circuit C that generates a unitary U . The answer is at least 4^n which is very big.

2.2 Deutsch's Algorithm

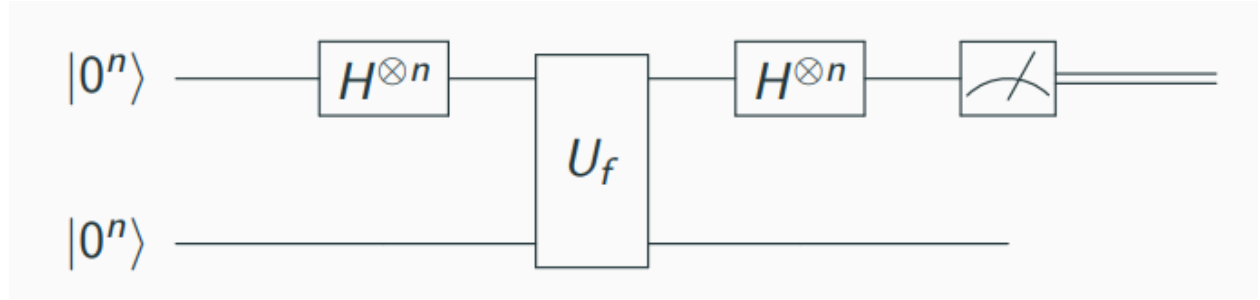
Given an oracle to a boolean function $f: (0,1) \rightarrow (0,1)$, it will decide whether $f(0) = f(1)$ or $f(0) \neq f(1)$

Claim Any classical algorithm that solves the Deutsch's problem must make 2 queries to f . For a quantum algorithm that can access a black box function f through a unitary U_f corresponding to the reversible version of f . That is

$$U_f|x, b\rangle = |x, b \oplus f(x)\rangle$$

The quantum algorithm solves the problem with only one call to U_f

$$|\phi_1\rangle = |+\rangle + |-\rangle = \frac{1}{2}(|00\rangle + |10\rangle - |01\rangle - |11\rangle)$$



$$|\phi_2\rangle = \frac{1}{2}(U_f|00\rangle + U_f|10\rangle - U_f|01\rangle - U_f|11\rangle) = |+\rangle + |-\rangle$$

$$|\phi_3\rangle = |0\rangle \otimes |-\rangle$$

We can test for all other base cases we can see that all other 3 combinations will work too.

Question How did the answer get encoded into the first bit which represents the input of the oracle?

3 10/2/24 Simon's Algorithm

3.1 Simon's Algorithm

We can have a classical algorithm that samples random inputs $x_1, \dots, x_k \in (0, 1)^n$, $K = 10\sqrt{2^n}$. We then evaluate the pairs for when $x_i \neq x_j$ where $f(x_i) = f(x_j)$ so the output is then $s = x_i \oplus x_j$. Using the **birthday paradox**, this algorithm will find the secret with $O(2^{n/2})$ queries to f .

For a quantum algorithm, we then call $2n$ qubits through a unitary U_f that maps $|x, z\rangle \rightarrow |x, z \oplus f(x)\rangle$. This is a classical-quantum hybrid algorithm that utilizes a quantum computer as a subroutine to sample from a distribution and uses classical post-processing.

Subroutine It queries the black box only once and there will be a string y such that the inner product of y and secret string s is equal to 0

$$s \cdot y = s_1 y_1 \oplus s_2 y_2 \oplus \dots \oplus s_n y_n$$

For the classical post processing, we obtain $m = 100n$ samples of $y^{(n)}$ such that $y^{(1)} \cdot s = 0, y^{(2)} \cdot s = 0, \dots, y^{(n)} \cdot s = 0$. With high probability, we can solve the system of equation and get s .

We know that

$$H^{\otimes n}|0\rangle^{\otimes n} = |+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in (0,1)^n} |x_1, \dots, x_n\rangle$$

For

$$\begin{aligned} H^{\otimes n}|x_1, \dots, x_n\rangle &= \frac{1}{\sqrt{2^n}}(|0\rangle + (-1)^{x_1}|1\rangle \otimes (|0\rangle + (-1)^{x_2}|1\rangle) \otimes \dots \otimes \\ &= \frac{1}{\sqrt{2^n}} \sum_{y_1, y_2, \dots, y_n \in (0,1)} (-1)^{x_1 y_1} |y_1\rangle \dots (-1)^{x_n y_n} |x_n y_n\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in (0,1)^n} (-1)^{x \cdot y} |y\rangle \end{aligned}$$

For the circuit, we have the following:

$$|\phi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x_1, \dots, x_n\rangle \otimes |0\dots 0\rangle$$

$$|\phi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle \otimes |f(x)\rangle$$

$$|\phi_3\rangle = H^{\otimes n} \otimes \mathbf{I}^{\otimes n} |\phi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_x H^{\otimes n} |x\rangle \otimes |f(x)\rangle$$

WLOG, we can assume we measured the 2nd register before applying the Hadamard gates to the 1st register of qubits. We see that this measurement at $|\phi_2\rangle$ leads us to determine the collapses of the function such that we only have two results $|x_i\rangle$ and $|x_j\rangle$.

After measuring the 2nd register, we obtain $z \in \{0,1\}^n$ in range (f) . Since f is a two to one function, the input register will collapse into one of two states $\implies x_i \oplus x_j = s$. Thus we get that

$$\begin{aligned} |\phi_3\rangle &= \frac{1}{\sqrt{2}} (H^{\otimes n} |x\rangle + H^{\otimes n} |u\rangle) \otimes |z\rangle \\ &= \frac{1}{\sqrt{2^{n+1}}} \left(\sum_y ((-1)^{x \cdot y} + (-1)^{u \cdot y}) |y\rangle \right) \otimes |z\rangle \end{aligned}$$

$u = x \oplus s$. Thus we see that when we have the following $(-1)^{x \cdot y} (1 + (-1)^{s \cdot y})$. If $s \cdot y = 0 \implies 2(-1)^{x \cdot y}$. If $s \cdot y = 1 \implies 0$

4 Grover's Algorithm

There exists exponential speedups in both Simon's algorithm and order finding/factoring which are used for highly structured problems. These problems all use a version of the quantum Fourier transform. Let's say that instead doing an unstructured problem that takes $f: \{0,1\}^n \rightarrow \{0,1\}$. We simply want a market input for x such that $f(x) = 1$ if it exists. Classically this takes $\Omega(2^n)$ which is choosing randomly and must look over all possible inputs before finding the correct one. This is very similar to searching through a database in which Grover can do. Grover's algorithm is a quadratic speedup which helps solve many problems.

- Simons and Deutsch algorithms called the oracle via the XOR oracle s.t. $U_f |x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$
- Grover's uses an oracle via the phase oracle s.t. $O_f(x) |x\rangle = (-1)^{f(x)} |x\rangle$ which is a reversible unitary operation and in principle can be applied. For Grover, this is more convenient. This becomes a relative phase after being put in superposition.
- These two oracles are equivalent in power where each one can be converted to do the other.

XOR Oracle $\implies$ Phase Oracle. $U_f |x, b\rangle \rightarrow O_f |x, b\rangle$ We can convert this using Deutsch's algorithm by performing a phase flip that maps from $b \oplus f(x)$ to $f(x) + \overline{f(x)}$. We can apply a z gate to the last gate apply $U_f^\dagger$ s.t. $U_f^\dagger Z U_f |x, b\rangle$. Additionally we can simply by H on the ancilla qubit thus we get $|-\rangle$ state which we can rewrite as $\frac{1}{\sqrt{2}}(f(x) - \overline{f(x)})$. Thus we will get

either the $|-\rangle$ state or the negative of the $|-\rangle$ state. Thus, we have picked up our phase through $(|0\rangle^n \otimes H|1\rangle)U_f(I^n \otimes H)|x, b\rangle$

If we wish to write a controlled O_f gate, we want to simulate the XOR oracle in the Hadamard basis. We can start in the $|+\rangle$ state apply the CU_f gate and apply another Hadamard. This will get us $\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{f(x)}|1\rangle)|x\rangle$ Thus in the case where $b = 0$, we will get $|+\rangle|f(x)\rangle$ or $-|-\rangle|f(x)\rangle$

- Assume there is exactly one marked input, Grover's circuit applied $H^{\otimes n}O_fR$ done $\sqrt{N}$ times which is also known as the Grover iterate
- The R unitary is defined to be $2|+\rangle\langle+|^{\otimes n} - I$ This is unitary and does a reflection through the all $|+\rangle$ state. This can be seen if we write $|\psi\rangle = \alpha|+\rangle + \beta|\psi^\perp\rangle$. The Grover diffusion operator $R|\psi\rangle = (2|+\rangle\langle+|^{\otimes n} - I)|\psi\rangle$. Continuing this derivation we will get $\alpha|+\rangle^{\otimes n} - \beta|\psi^\perp\rangle$
- Let x^* denote the marked input, the intermediate states of Grover's is a superposition of $|x^*\rangle$ and Δ where Δ are just values from a superposition.

Proof By Induction The base case is going to be for one marked item such that we have a linear combination between $|\Delta\rangle + |x^*\rangle$ Claim that both O_f and R are linear combinations of $|\Delta\rangle, |x^*\rangle$. Applying to $O_f|\psi\rangle = \alpha O_f|\Delta\rangle - \beta|x^*\rangle$, this will be true for R too. $R|\psi\rangle = \alpha R|\Delta\rangle + \beta R|x^*\rangle$. $R|\Delta\rangle = (2|+\rangle\langle+|^{\otimes n} - I)\frac{1}{\sqrt{2^n-1}}\sum_{x \neq x^*}|x\rangle$. We can write $|\Delta\rangle$ as $|+\rangle$ where $|+\rangle^{\otimes n} = \sqrt{\frac{2^n-1}{2^n}}|\Delta\rangle + \frac{1}{\sqrt{2^n}}|x^*\rangle$ collecting the correct terms and doing the algebra we will get $= \sqrt{\frac{2^n}{2^n-1}}|+\rangle^{\otimes n} - \frac{1}{\sqrt{2^n-1}}|+\rangle^{\otimes n} \dots$ and by proof of hand-waviness and intimidation, it will stay in the two dimensional subspace.

Going back to this proof, we can rather see that $R|\psi\rangle = (2|\psi\rangle\langle\psi|^{\otimes n} - I)|\psi\rangle$ and we see that this is already a linear combination that is properly shown from the above.

- We see that for one run our chance of finding $|x^*\rangle$ we will have a chance of $\frac{1}{\sqrt{2^n}}$ chance of finding our value
- as we repeat this process running Grover's subroutine, we will have a high change after running a total of $\sqrt{2^n}$ times in order to get the probability.
- The angle θ between that of $|+\rangle$ and $|\Delta\rangle$ will continue in a linear format s.t. we have $\sqrt{2^n}\theta$
- After k iterations of O_f , then we will have that the k th state of Grover's is going to be of the form $\cos((2k+1)\theta)|\Delta\rangle + \sin((2k+1)\theta)|x^*\rangle$. Solving we wish to have that $\sin((2k+1)\theta) \approx 1 \iff (2k+1)\theta \approx \frac{\pi}{2}$
- Thus we see a sinusoidal behavior behind Grover's algorithm in which the success probability is periodic. Thus, the runtime should be ensured s.t. the number of iterations is $\sqrt{2^n}$ times.

What happens if there are multiple solutions to the search problem? For multiple solutions, the states are still restricted by a 2 dimensional subspaces which are $\text{span}[\text{solutions}, \text{non-solutions}]$ and the number of iterations to get a solution will depend on the number of solutions of $\mathcal{O}(\sqrt{N/M})$ where M are the number of solutions and will find only one solution. Thus, given that M is unknown, then a number must be picked between $\sqrt{N}$ to 1 s.t. a solution can be found 50% of the time yielding a $O(1)$ speedup. This can be done with different number of iterations $T_0, \lambda T_0, \lambda^2 T_0$ This will allow us to find a solution quite quickly where $1 < \lambda < \frac{4}{3}$.

- For database search, we can encode $f(x)$ to be an if statement to output 1 given a set of truths for that person. The algorithmic complexity of running this to compute $f(x)$ could be much larger and could lead to a problem with the black box itself.
- A better use is where computing $f(x) < N$. For example, finding a SAT problem is a NP complete problem where a SAT problem is (the problem of determining if there exists an interpretation that satisfies a given Boolean formula). With Grover, we can solve it in $O(\sqrt{2^n})\text{poly}(n)$ time

5 Grover's Analysis

Note that $|\psi\rangle = \cos((2k+1)\theta)|\Delta\rangle + \sin((2k+1)\theta)|x^*\rangle$ where $\theta = \sin^{-1} \frac{1}{\sqrt{N}}$. We take an inductive step to assume this is true for $k \geq 1$ Then apply one more grover iteration. We see that when applying $RO_f|\psi\rangle$ we will get the correct values for increasing θ . We can rewrite $R|\Delta\rangle = \cos(2\theta)|\Delta\rangle + \sin(2\theta)|x^*\rangle$ which gives us the correct term. Similarly, $R|x^*\rangle = \sin(2\theta)|\Delta\rangle + \cos(2\theta)|x^*\rangle$. Thus we get the desired result that we wished for initially s.t.

$$RO_f|\psi\rangle = \cos((2k+3)\theta)|\Delta\rangle + \sin((2k+3)\theta)|x^*\rangle$$

For multiple solutions s.t. $M > 1$, we can find a solution in $O(\sqrt{N/M})$ queries where our state is very close to an intermediate state which is a solutions since the state of solutions is $|\Gamma\rangle = \frac{1}{\sqrt{M}}$. What if now want to find all solutions? There are $O(\sqrt{N/M})$ query solutions to this.

- Use $\frac{\sqrt{N}}{\sqrt{M}}$ to find the first solution
- Grover search will then be updates s.t. the solution x_1 is excluded and can find x_2 with a solution space $\sqrt{\frac{N}{M-1}}$
- The total number of queries is going to be

$$\sum_{j=0}^{M-1} \sqrt{\frac{N}{M-j}} \leq O(\sqrt{MN})$$

5.1 Quantum Counting

6 Quantum Complexity Theory

In computational complexity theory, a complexity class is a set of computational problems "of related resource-based complexity". The two most commonly analyzed resources are time and memory.

The discussion of decision problems whose solutions can be verified in polynomial time shows it is provable. Examples include Boolean satisfiability, and travelings salesman. Most optimization/search problems are NP and NP-complete.

- For bounded error probabilistic polynomial time, the randomized solution generator should be right more than 99% of the time. It is conjectured that $P = BPP$.
- PSPACE: polynomial space where the algorithm is space bounded by a small amount of memory.

Thus, there exists problems which can be solved without considering the space complexity or n^c . Additionally in $P_{time} \implies P_{space}$

- Decision problems that can be solved by a deterministic algorithm are called exponential time such as $O(2^{n^c})$ time. This is for all of $PSPACE$ and works for chess too.

Decision problems that can be solved by a quantum algorithm are called BQP s and are of size $O(n^c)$ with a high probability of solution using this. We see that factoring is in BQP . Other examples include simulations in quantum physics. We know that BQP is not known to be in $PSPACE$, NP , or P . For the following notes, we will see how BQP is part of P_{space} . The central questions are thus $BQP \subseteq BPP$, $NP \subseteq BQP$. How can we show that this is true?

Definition We define a problem called the circuit acceptance probability problem. If this is done in the complexity class, then it can be done in BQP . Thus, given a quantum circuit $C|\psi\rangle$ and we can check the first qubit to yield $|1\rangle$ would be close to 1. Thus we have an input of C and an output that determines the case of whether or not $|0\dots 0\rangle$ has 0 or 1 probability. The $APPROX - Q - Circuit$ is the canonical BQP complete problem which implies it is the hardest problem in BQP . Thus, if this is solvable than all other problems are BQP .

Claim $APPROX-Q-CIRCUIT$ is solvable in BQP . Proof: The solution is tautological

6.1 Bounding AQC

We can show that $APPROX-Q-CIRCUIT$ or $AQC \in EXP$ based on applying gates to it.

From here we wish to show that it is also in P_{space} . TO do this, we use a sum-over-histories or **Feynman Path Integral**. The key idea is that we can iterate over exponentially many paths of quantum computation to find the final probability. Doing this is space efficient by calculating the one amplitude and saving it and thus there is no reason to store multiple strings anymore and their computation at the same time.

1. root node is labeled $|00\dots 00\rangle$
2. each node has 2^n children
3. edge from node $|x\rangle$ in layer t to node y in the next layer is labeled by a transition amplitude called $\langle y|g_t|x\rangle$

Claim: Given n -qubit basis $|x\rangle |y\rangle$ and two qubit gate g , the transition can be calculated in polynomial time Proof: Assume WLOG, g acts on the first two qubits we can calculate that g is a 4×4 unitary which implies that this is polynomial time.

Subroutine: Compute Amp The amplitude to calculate the basis states x and y are done by calculating the intermediate steps $u_1, u_2, \dots, u_{T-1}$ where we get that $amp = \langle y|g_T|u_{T-1}\rangle \langle u_{T-1}|g_{T-1}|u_{T-1}\rangle \dots \langle u_1|g_1|x\rangle$

This subroutine computes the transition in $polyn$ time to store amp .

7 Quantum Error Correction

The quantum errors can be digitized and can reduce it to a discrete set of errors. There exists the following:

- Bitflip error with the X gate
- phaseflip error with the Z gate
- bitflip and phaseflip error with XZ gate

We can use the repetition code and we can copy a specific basis qutite easily but not a state. This is done by taking $|0\rangle \implies |000\rangle = |\bar{0}\rangle$. Doing this with the state $|\psi\rangle$ will get us $|\bar{\psi}\rangle$. Suppose there exists a bit flip error that flips the middle qubit. How do we measure such an error without collapsing the state? Instead, there is a computation on ancilla qubits which are the error syndromes.

Bit flip error

- Suppose we have state $|\bar{\psi}\rangle$
- The error is factored out and not part of the superposition state. Thus, the syndrome qubits don't effect the computation
- Suppose we have the state $|\bar{\psi}\rangle$ and we can do the error correction procedure with $CNOT$ gates. Thus we can create a table with the different syndromes. There exists the possibility of a and b thus we will know all possible formations of an error residing in the qubits
- The correction procedure is simply applying the $CNOT$ gate back onto the quantum state and thus protects against possible sources of error.

Phase flip error

- Apply the hadamard's to the whole basis and apply a bit flip error correction code. To recover the state apply the Hadamard basis again after applying the error correction recovery.

8 Shor's Code

$$|\bar{\psi}\rangle = \alpha(|000\rangle + |111\rangle)^{\otimes 3} + \beta(|000\rangle - |111\rangle)^{\otimes 3}$$

. Suppose a bit is flipped during this error correcting code here on the middle qubit of $\otimes 2$ or rather $|\tilde{\psi}\rangle = X_5|\psi\rangle$. The correction and decoding procedure consists of a correction phase and a decoding phase where the error is detected and then handled.